CYBERSECURITY INTERN REPORT AT SHADOWFOX

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Task Level: Beginner & Intermediate Level

Task Level (Beginner & Intermediate level)

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Task Level (Beginner)

➢ Introduction:-

In the domain of cybersecurity, it is vital to understand and detect vulnerabilities in web applications to protect systems from potential cyber threats. Web applications are frequently targeted by attackers due to misconfigurations, insecure communication, and exposed services.

During my virtual internship, I was assigned the responsibility of conducting basic security evaluations on the vulnerable test website:

http://testphp.vulnweb.com/

➢ Information about the Report:-

The report consists of three major security assessment tasks performed on the given website. Each task focuses on a different aspect of web application security.

1. Port Scanning

Port scanning was the initial task performed to identify all open ports on the target website. This technique is a basic reconnaissance method used to determine which network ports are active and what services are running on them. The purpose of this task was to identify exposed or unnecessary ports and understand possible entry points that attackers could exploit to gain unauthorized access to the web server.

2. Brute Forcing Directories on Website

The second task involved performing directory brute forcing on the website to identify accessible directories that are not directly visible to users. Many web applications contain hidden or unlinked directories that may be insecure or poorly configured. This task helped in identifying unprotected directories and potential paths that could expose sensitive files or confidential information if exploited by attackers.

3. Network Traffic Interception

Finally, network traffic interception was performed by logging into the website and capturing network packets using Wireshark. This task focused on analyzing the data exchanged between the client and the server in order to observe how information is transmitted during the login process and to identify any sensitive data that may be exposed due to insecure communication.

- **Required Machine:-**

- ✓ Standard computer system with reliable network connectivity.
- ✓ Minimum 4 GB RAM (8 GB recommended for smooth performance).

Task 1 : Port Scanning

- To Find the ports that are open on the website http://testphp.vulnweb.com/

➢ **Attack Name:-**

Port Scanning (Network Reconnaissance)

➢ **Severity:-**

Low to Medium

Reason for Severity:

Port scanning itself does not damage the system.However, it reveals open ports and running services.This information can be used by attackers for further exploitation.Hence, it is considered a reconnaissance-level threat.

➢ **Steps Performed with screenshots:-**

1) The domain name **testphp.vulnweb.com** was first resolved to its IP address using the dig command.
2) The DNS query returned the IP address **44.228.249.3** for the target website.

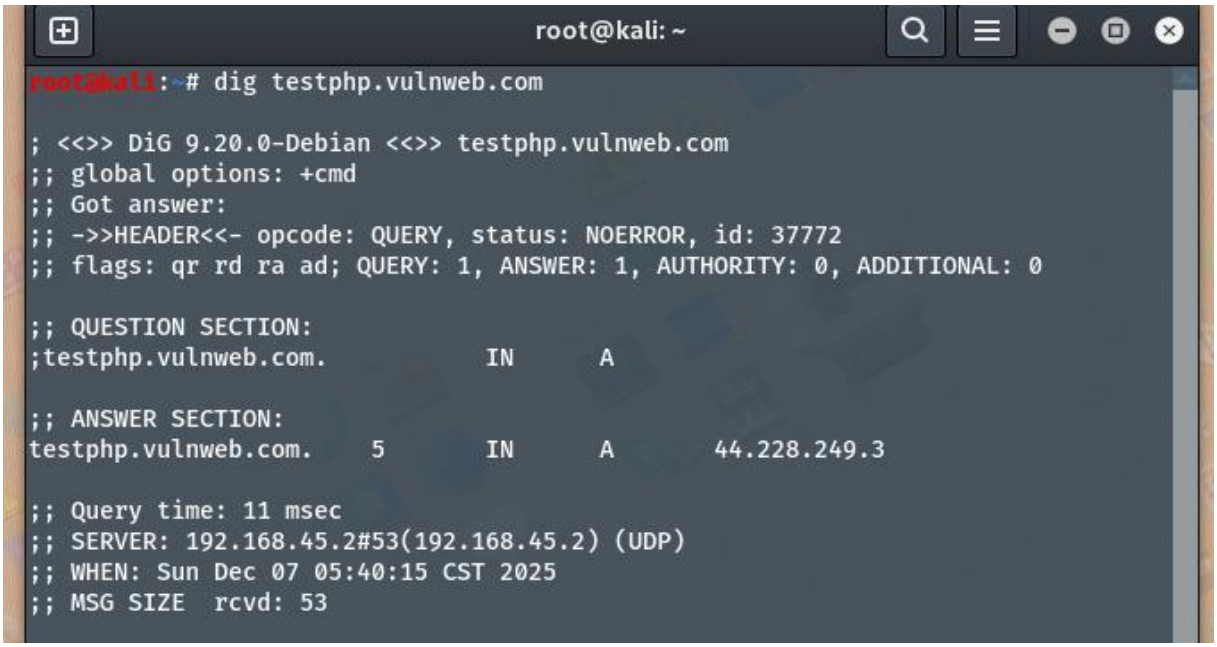

Figure 1.1: DNS resolution using dig command.

3) After identifying the IP address, a port scan was performed using the nmap tool.
4) The scan was executed against the resolved IP address to identify open TCP ports.

Figure 1.2: Nmap Scanning

5) The scan result showed that **port 80 (HTTP)** was open on the target server.
6) Other ports were found to be filtered or not responding.

➢ **Analysis:-**

- The DNS lookup confirmed that the website is hosted on an Amazon AWS server.
- The port scanning result revealed that the web server is accessible over **HTTP (port 80)**.
- No additional services were exposed, which reduces the attack surface.
- However, the presence of an open HTTP port indicates that communication may not be encrypted.

➢ Impact:-

Advantages (for Security Testing)

1. Helps identify exposed services on the server
2. Assists administrators in understanding network configuration
3. Acts as a first step in vulnerability assessment

Disadvantages (Security Risks)

4. Open ports can attract attackers
5. Attackers can use this information for targeted attacks
6. Lack of HTTPS may lead to data interception risks

➢ Mitigation Steps:-

- Close all unnecessary ports on the server
- Use firewalls to restrict access to required ports only
- Enable HTTPS (TLS/SSL) instead of plain HTTP
- Monitor and log port scanning attempts
- Use intrusion detection and prevention systems (IDS/IPS)

➢ Resources Used:-

✧ Operating System: Kali Linux
✧ Tools Used**:**
✧ dig – for DNS resolution
✧ nmap – for port scanning
✧ Target Website: http://testphp.vulnweb.com/
✧ Internet Connection

Task 2: Directory Brute Forcing

- Brute force the website http://testphp.vulnweb.com/ and find the directories that are present in the website.

➢ Attack Name:-

Directory Brute Forcing (Directory Enumeration)

➢ Severity:-

Score: 4.3

Severity Level: Medium

➢ Steps Performed:-

1) The target website **http://testphp.vulnweb.com/** was selected for testing.
2) Gobuster was executed in directory enumeration mode.
3) A predefined common wordlist was used to guess directory names.
4) Multiple HTTP GET requests were sent to the web server.Server responses were analyzed using HTTP status codes.

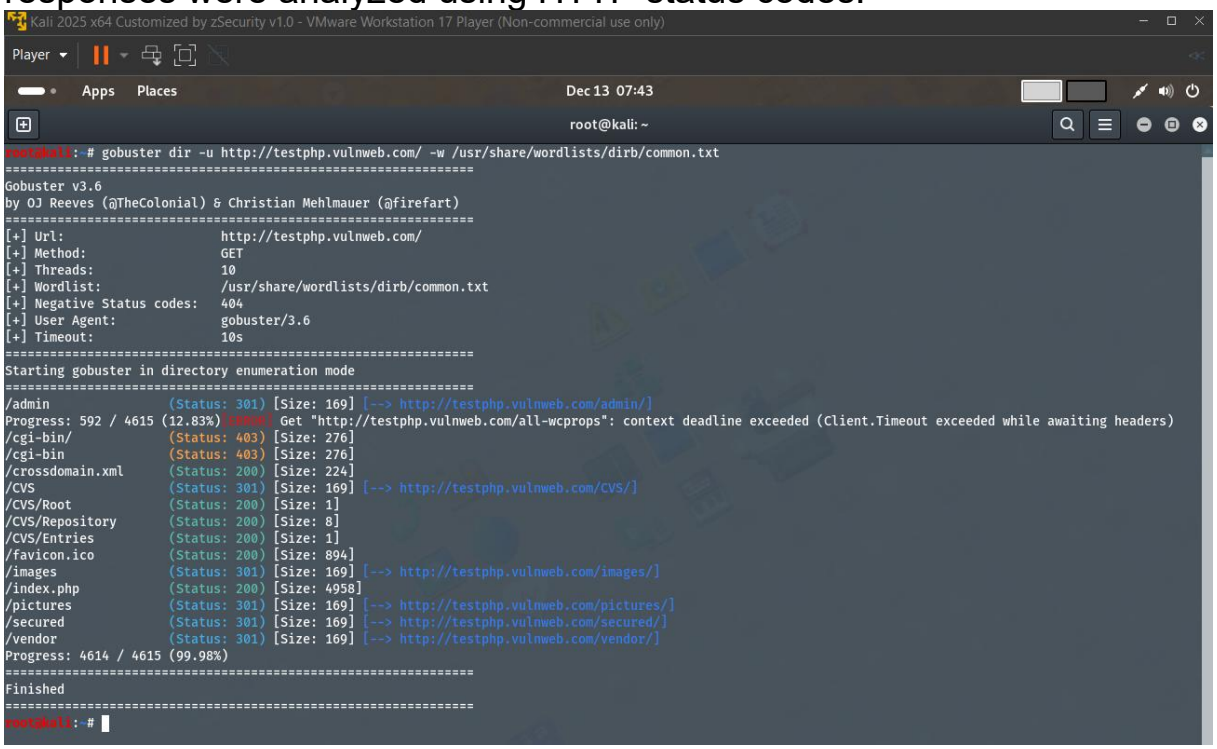

Figure 2: Gobuster scanning

5) Valid directories such as /admin, /images, /secured, /vendor, and /pictures were identified.
6) The scan completed successfully and displayed accessible directories.

➢ **Analysis:-**

- The results show that several directories are accessible on the website.
- Some directories returned **301 (redirect)** responses, indicating active paths.
- The presence of directories like /admin and /secured suggests potential sensitive areas.
- Lack of proper access restrictions increases the risk of unauthorized access.
- This indicates weak directory access control on the web server.

➢ **Impact:-**

1. Exposes hidden or unlinked directories on the web server
2. Reveals sensitive paths such as admin panels or configuration folders
3. Can assist attackers in planning further attacks
4. Increases the attack surface of the web application

➢ **Mitigation Steps:-**

- Restrict access to sensitive directories using authentication
- Remove unused or test directories from the server
- Disable directory listing in the web server configuration
- Implement proper access control policies
- Use a Web Application Firewall (WAF)
- Perform regular security assessments

- **Resources Used:-**

- Operating System**:** Kali Linux
- Tool Used**:** Gobuster
- Wordlist**:** /usr/share/wordlists/dirb/common.txt
- Target Website**:** http://testphp.vulnweb.com/

Task 3: Network Traffic Interception

- **Attack Name:-**

Network Traffic Interception (Credential Sniffing over HTTP)

- **Severity:-**

Score: 7.4

Severity Level: High

Reason for Severity:-

User credentials are transmitted in **plain text.**No encryption is used during login.Attackers can easily capture usernames and passwords.Direct compromise of user accounts is possible.

- **Steps Performed with screenshots:-**

1) The target website login page http://testphp.vulnweb.com/login.php was opened in the browser.
2) A login attempt was made using valid credentials.
3) Wireshark was started on the active network interface (eth0).
4) Network traffic was captured during the login process.

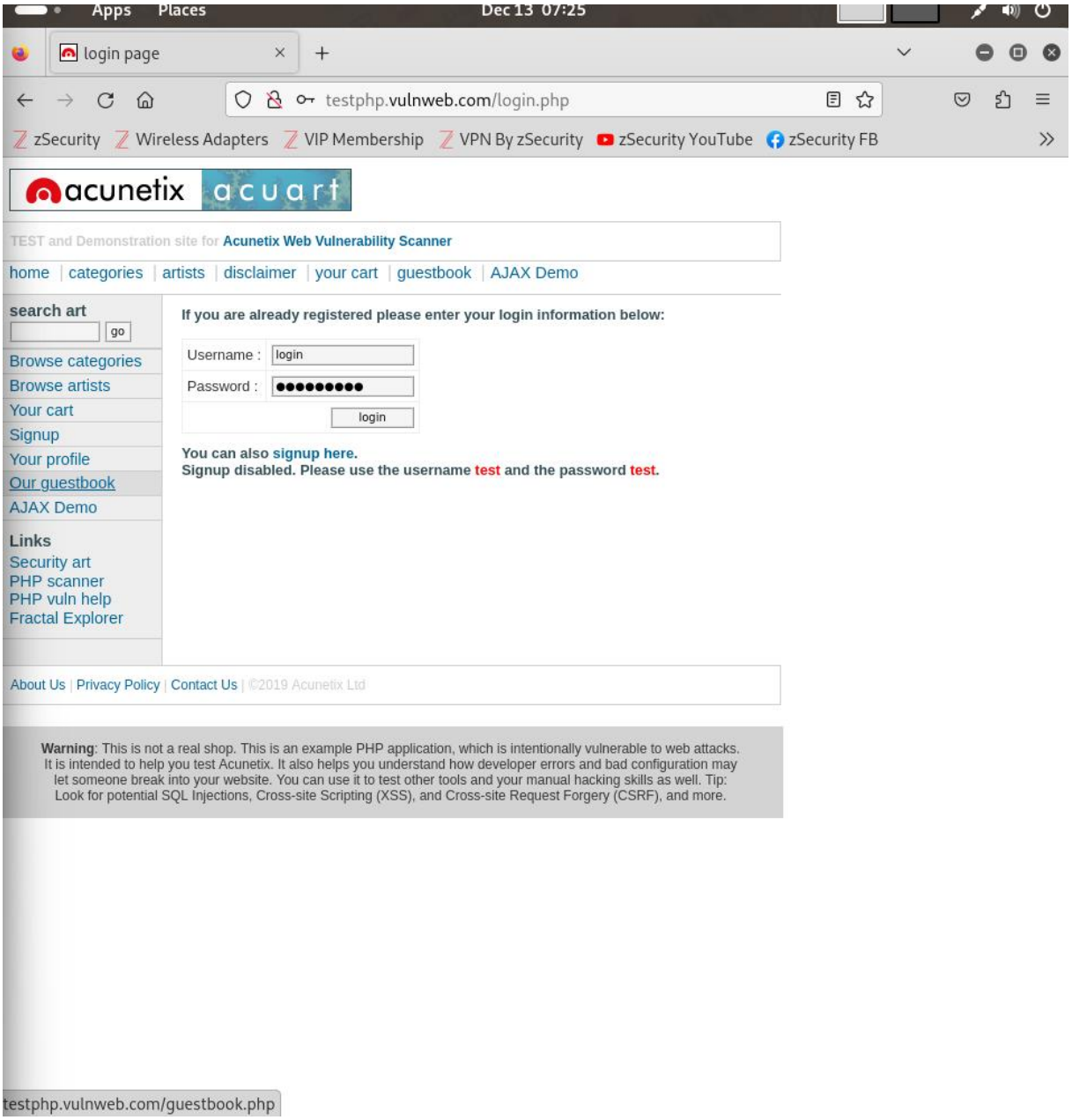

Figure 3.1:Login attempt

5) HTTP filter was applied to analyze only HTTP packets.
6) The POST request containing login data was identified.

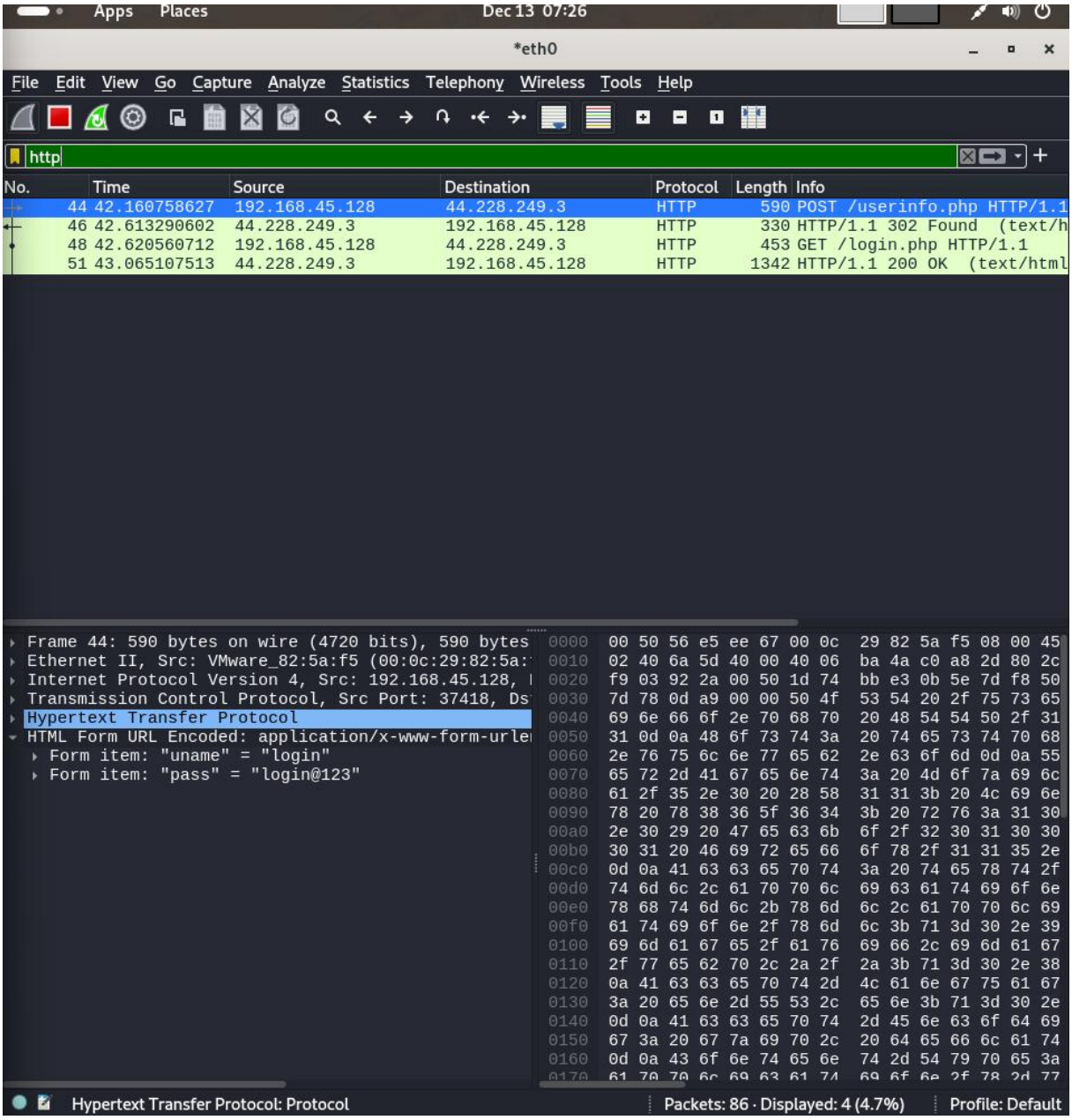

Figure 3.2: Wireshark result

7) Username and password were visible in plain text within the captured packets.

➢ Analysis:-

- The captured packets show that the website uses **HTTP instead of HTTPS**.
- Username and password fields are clearly visible in the packet payload.

- Any attacker on the same network can intercept and read sensitive information.
- This indicates a serious lack of secure communication mechanisms.

➢ **Impact:-**

1. Exposure of usernames and passwords
2. Unauthorized access to user accounts
3. Possibility of account takeover
4. Loss of confidentiality of sensitive data
5. Increased risk of further attacks using stolen credentials

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➢ **Mitigation Steps:-**

- Implement HTTPS using SSL/TLS certificates
- Redirect all HTTP traffic to HTTPS
- Encrypt sensitive data before transmissio
- Use secure authentication mechanisms
- Implement HTTP Strict Transport Security (HSTS)
- Regularly test network traffic for sensitive data exposure

➢ **Resources Used:-**

✧ Operating System**:** Kali Linux
✧ Tool Used**:** Wireshark
✧ Protocol Analyzed**:** HTTP
✧ Network Interface**:** eth0
✧ Target Website: http://testphp.vulnweb.com/
✧ Web Browser: Firefox

Task Level (Intermediate)

- **Introduction:-**

As cybersecurity threats continue to evolve, it is essential for security professionals to possess hands-on experience with real-world attack and defense techniques. Intermediate-level cybersecurity tasks focus on strengthening practical skills related to encryption analysis, malware execution flow, exploitation frameworks, and wireless network security.

During this virtual internship, I was assigned a set of intermediate-level tasks designed to enhance my understanding of system security, digital forensics, penetration testing, and wireless security. These tasks involve working with disk encryption tools, executable analysis, exploitation frameworks, and Wi-Fi security mechanisms. Each task was performed in a controlled and ethical environment for learning and assessment purposes only.

- **Information about the Report:-**

4. Decrypting a VeraCrypt Encrypted File

In this task, a file encrypted using **VeraCrypt**, a disk encryption tool, was provided. The password required to access the encrypted file was not given in plain text but instead provided in a hashed or encoded format inside a file named encoded.txt. The objective was to decode or crack the encrypted password, use it to unlock the VeraCrypt container, and retrieve the secret code stored inside the encrypted file.

5. Finding the Entry Point of a VeraCrypt Executable

In this task, an executable file of VeraCrypt was provided. The goal was to analyze the executable using a PE (Portable Executable) analysis tool

and identify the **entry point address** of the program. The entry point is the memory address where execution begins when the program is launched.

6. Creating a Reverse Shell Using Metasploit

This task involved creating a payload using the **Metasploit Framework** and establishing a reverse shell connection from a Windows 10 machine to the attacker machine within a virtual machine setup. The objective was to understand how attackers gain remote access to a system using exploitation frameworks.

7. Wi-Fi Deauthentication Attack and Password Cracking

In this task, a deauthentication (deauth) attack was performed on the user's own Wi-Fi network. The goal was to disconnect a device from the router, capture the WPA/WPA2 handshake, and then crack the Wi-Fi password using a custom wordlist that could contain the network password.

- **REQUIREMENT:-**

- ✓ Minimum 8 GB RAM (recommended)
- ✓ Wireless Adapter supporting Monitor Mode
- ✓ Kali Linux (Virtual Machine or Bare Metal)
- ✓ Windows 10 Virtual Machine
- ✓ VirtualBox or VMware Workstation
- ✓ VeraCrypt
- ✓ Metasploit Framework
- ✓ PE Explorer Tool
- ✓ Wireshark
- ✓ Aircrack-ng Suite

Task 4: Decrypting a VeraCrypt Encrypted File

- **Attack Name:-**

Offline Password Cracking on VeraCrypt Container (Hash Cracking)

- **Severity:-**

CVSS Score: 6.5

Severity Level: Medium

- **Steps Performed:-**

1) A VeraCrypt encrypted file was provided along with an encoded.txt file containing the encrypted password in hash format.
2) The encoded.txt file was accessed in Kali Linux to identify the hashed password.
3) The hash was analyzed and identified as an MD5 hash.The password hash was cracked using **John the Ripper** with the rockyou.txt wordlist.After successful cracking, the password password123 was obtained.

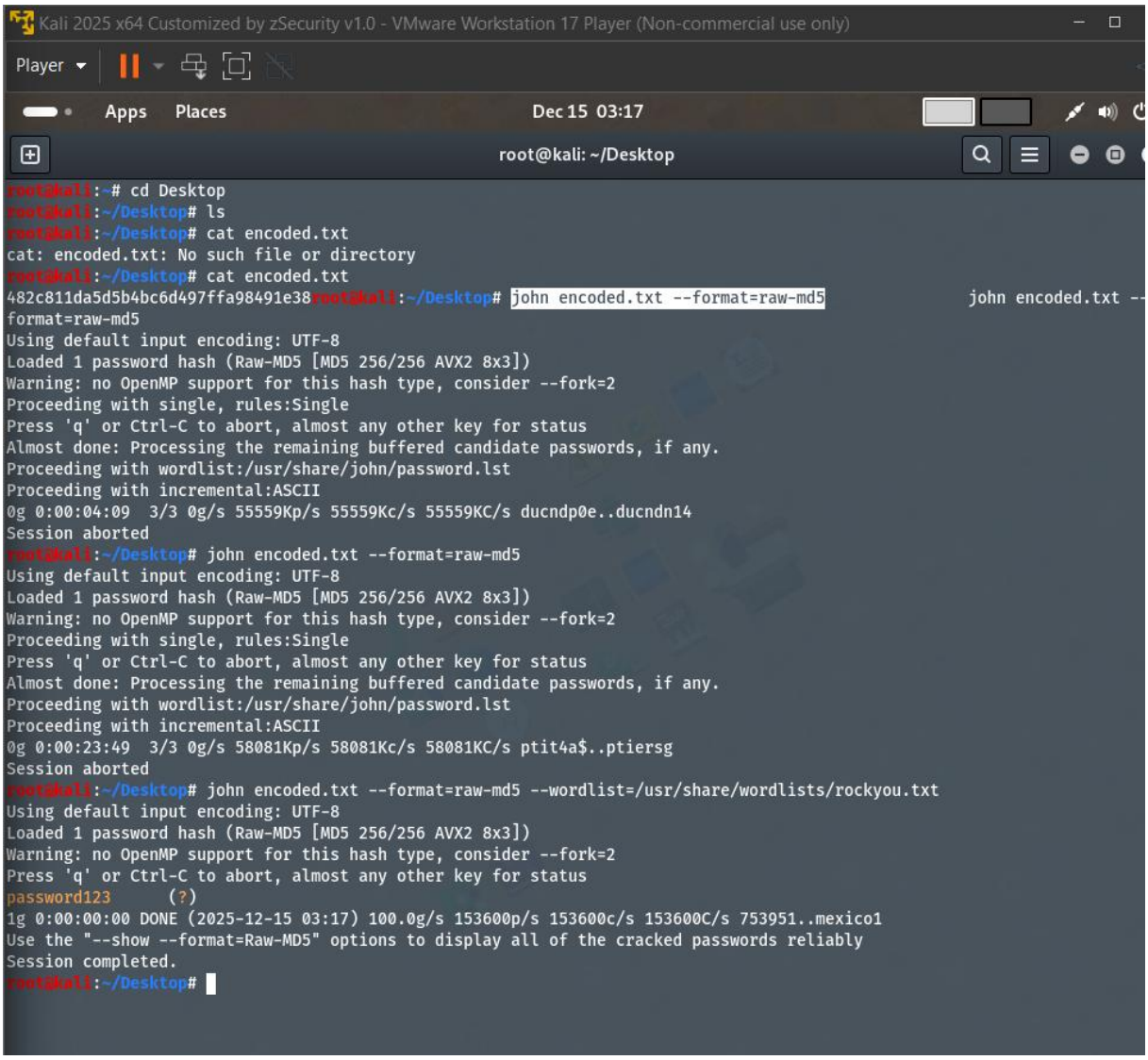

Figure 4.1:Password hash analysis using John the Ripper

4) VeraCrypt application was opened on the system.

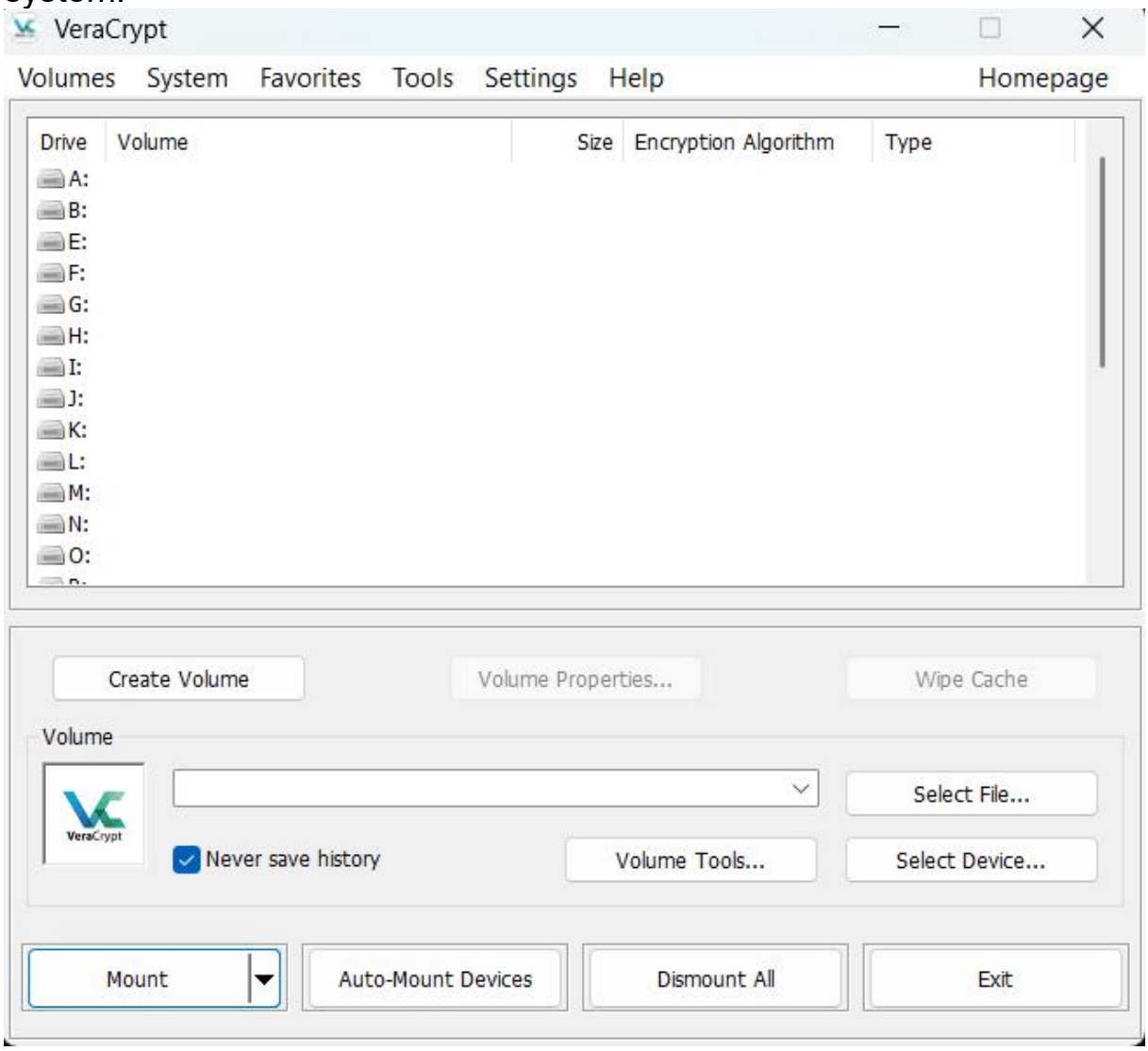

Figure 4.2:Veracrypt main interface

5) The encrypted file was selected and mounted using the recovered password.

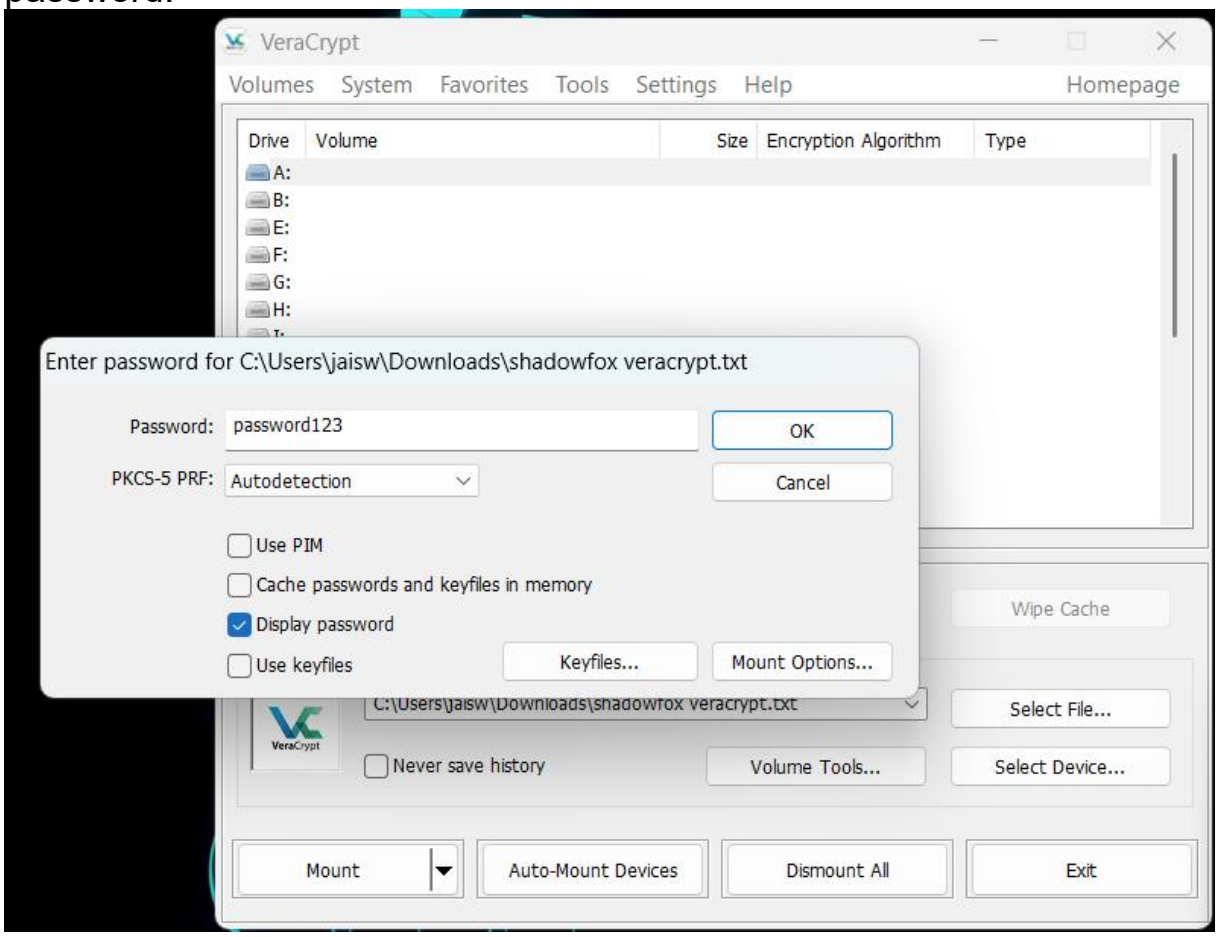

Figure 4.3:Password input for veracrypt volume

6) Upon successful decryption, the file contents were accessed.The secret code inside the decrypted file was found as:
“never giveup”

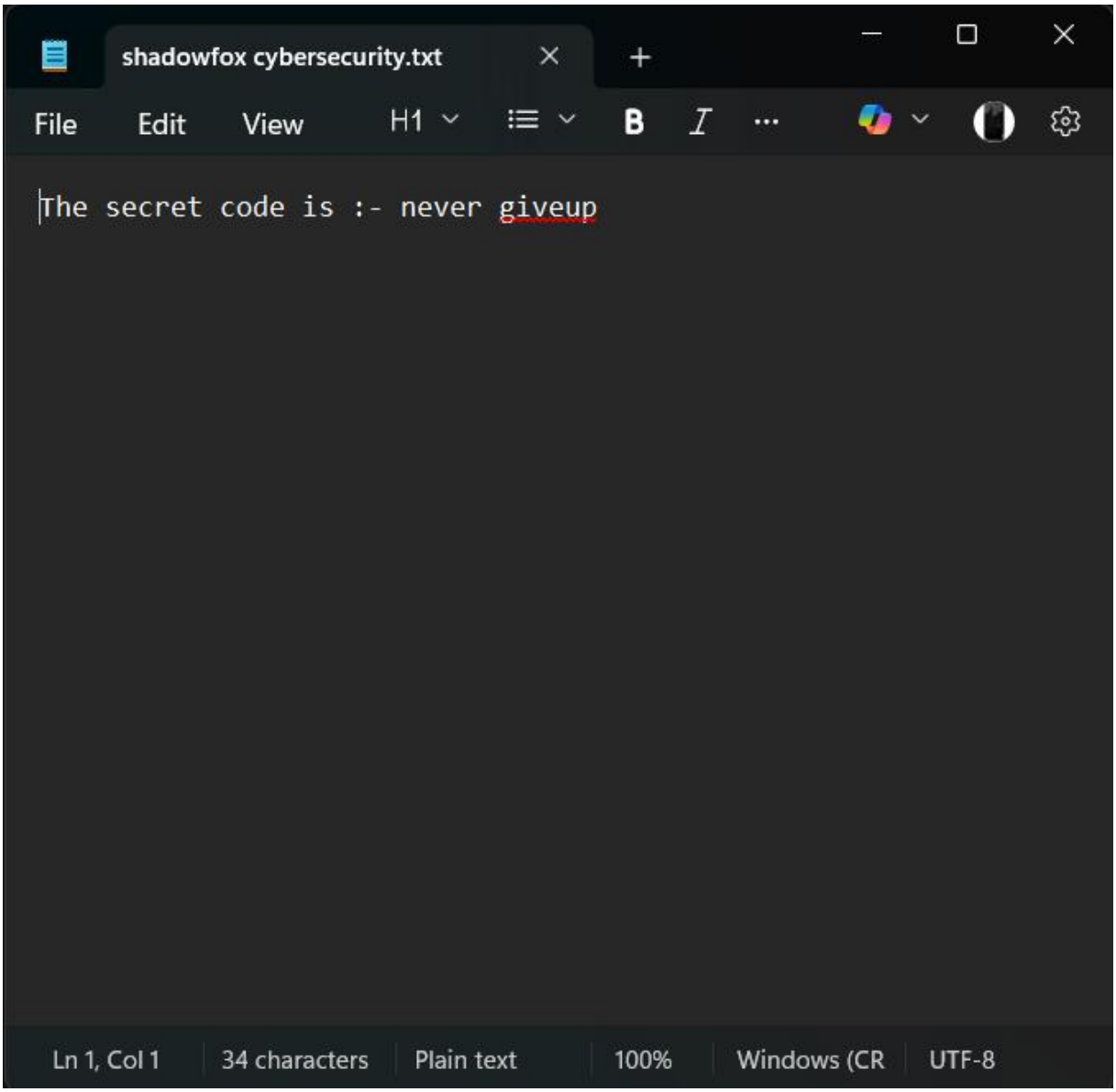

Figure 4.4:Extracted secret content

➢ **Analysis:-**

- The encrypted container was protected using a weak password.
- The password hash stored in encoded.txt was vulnerable to dictionary-based cracking.
- Use of common passwords significantly reduces the effectiveness of encryption.
- Although VeraCrypt is a strong encryption tool, weak password practices can compromise security.
- This task demonstrates how encryption security depends heavily on password strength.

➢ **Impact:-**

Advantages (From Security Perspective)

1. Demonstrates the importance of strong passwords in encrypted systems
2. Helps understand real-world risks related to poor password management
3. Useful for forensic investigations and password recovery scenarios

Disadvantages (Security Risks)

4. Weak passwords can allow attackers to decrypt sensitive data
5. Offline password cracking bypasses system-level protections
6. Confidential information can be exposed if encryption keys are compromised

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➢ **Mitigation Steps:-**

- Use strong and complex passwords for encrypted containers
- Avoid common or dictionary-based passwords
- Use keyfiles along with passwords in VeraCrypt
- Increase PIM (Personal Iterations Multiplier) value
- Store password hashes securely or avoid storing them altogether
- Educate users on secure password practices

➢ **Resources Used:-**

✧ **Operating System:** Kali Linux & Windows
✧ **Tools Used:**
✧ -VeraCrypt
✧ -John the Ripper
✧ **Wordlist:** rockyou.txt
✧ **File Provided:** encoded.txt
✧ **Target:** VeraCrypt encrypted container

TASK 5: Finding the Entry Point of a VeraCrypt Executable

- **Attack Name:-**

Executable Static Analysis (Entry Point Identification)

- **Severity:-**

CVSS Score: 4.0

Severity Level: Low

- **Steps Performed:-**

1) An executable file of VeraCrypt (VeraCrypt Setup 1.26.7.exe) was provided for analysis.
2) The PE Explorer tool was installed and opened on the Windows system.

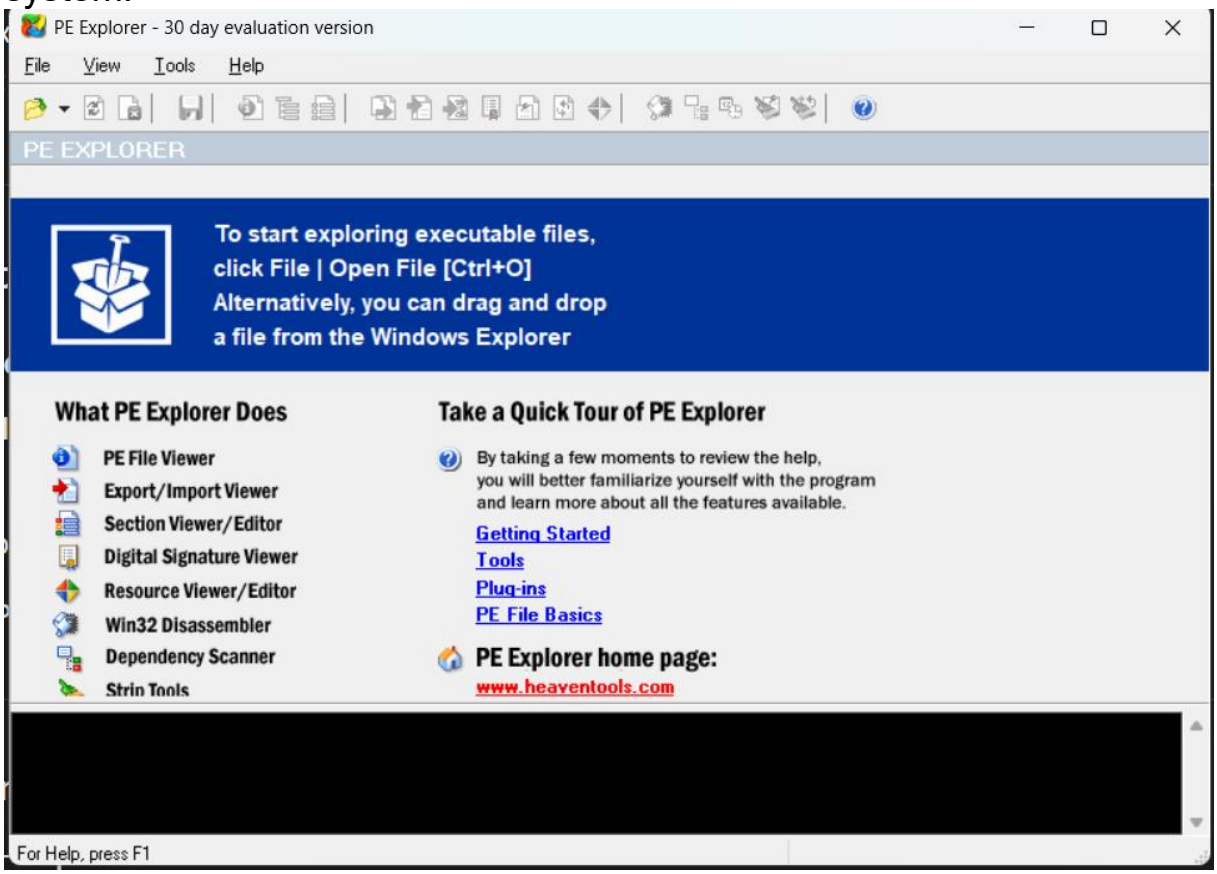

Figure 5.1:PE explorer interface

3) The executable file was loaded into PE Explorer using the *Open File* option.
4) The tool automatically analyzed the Portable Executable (PE) structure.
5) The **Headers Information** section was accessed.
6) The **Address of Entry Point (AEP)** field was located.
7) The entry point address was identified as **004237B0**.
8) A screenshot of the entry point value was captured as evidence.

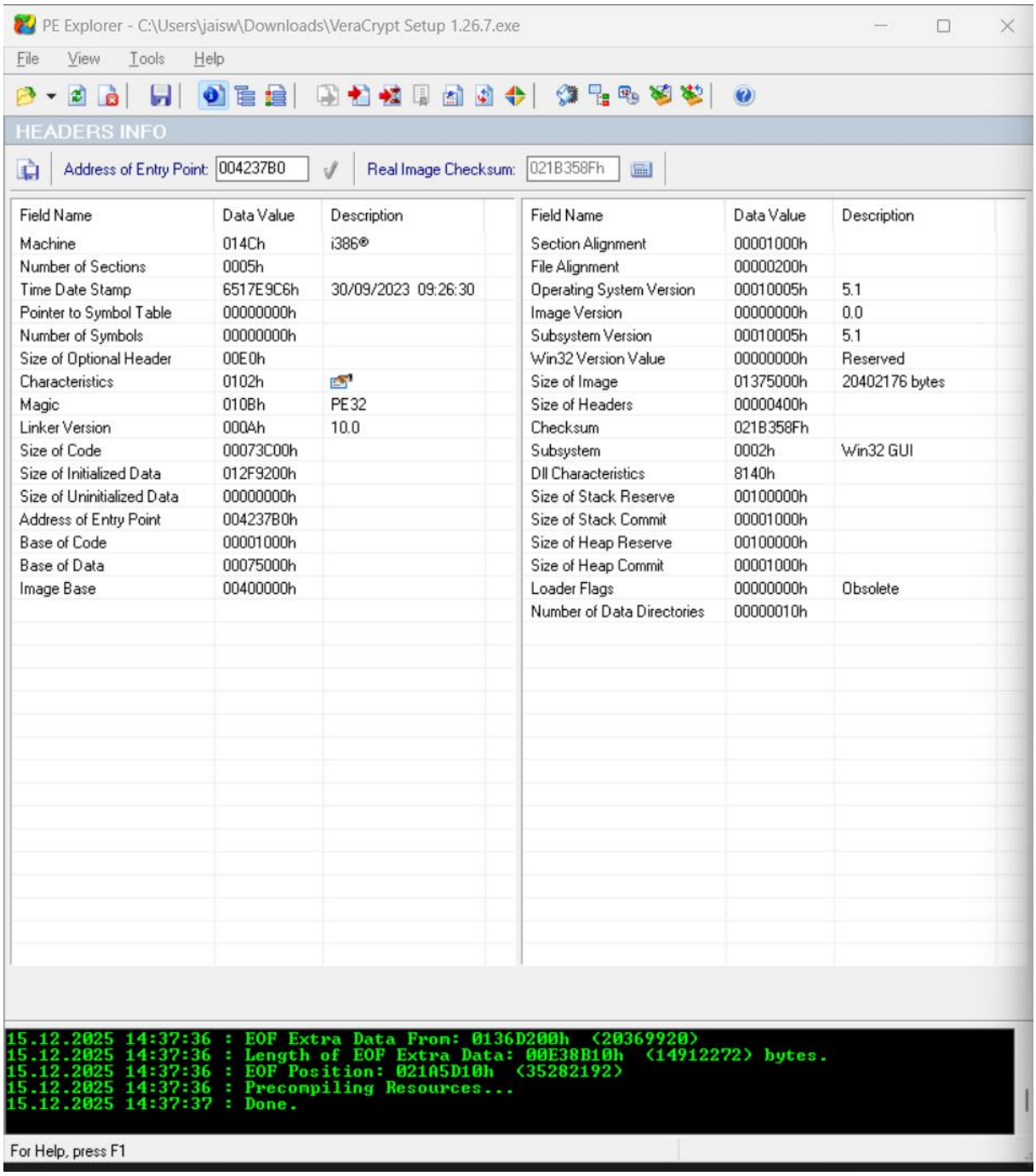

Figure 5.2:Entry point address

➢ Analysis:-

- The entry point represents the memory address where execution begins when the program is launched.
- Identifying the entry point is a fundamental step in static malware and executable analysis.
- This information is commonly used in reverse engineering and debugging.
- The VeraCrypt executable analyzed showed a valid and expected entry point, indicating a legitimate program structure.
- No obfuscation or packing was observed at the entry point level.

➢ Impact:-

Advantages (Security & Analysis Perspective)

1. Helps understand executable file structure
2. Useful for malware analysis and reverse engineering
3. Assists in debugging and program flow analysis
4. Important skill for digital forensics and SOC roles

Disadvantages (If Misused)

5. Attackers can use entry point information to inject malicious code
6. Can assist in creating malware or exploiting binaries
7. May aid reverse engineering of proprietary software

➢ Mitigation Steps:-

- Use code signing to verify executable integrity
- Implement executable packing or obfuscation where required
- Monitor unauthorized analysis or tampering of binaries
- Restrict access to sensitive executables
- Use anti-reverse-engineering techniques in critical applications

- **Resources Used:-**

- **Operating System:** Windows
- **Tool Used:** PE Explorer
- **Executable File:** VeraCrypt Setup 1.26.7.exe
- **Analysis Type:** Static Analysi

TASK 6: Creating a Reverse Shell Using Metasploit

- **Attack Name:-**

Reverse Shell Attack using Metasploit Framework

- **Severity:-**

CVSS Score: 8.8

Severity Level: High

- **Steps Performed:-**

1) A Kali Linux machine was used as the attacker system and a Windows 10 virtual machine was used as the target.
2) A reverse shell payload was created using the Metasploit msfvenom utility and the host and port number have been set.The payload was configured to connect back to the attacker's IP address and listening port.The payload was generated in executable format (.exe) and saved as simple_shell.exe.

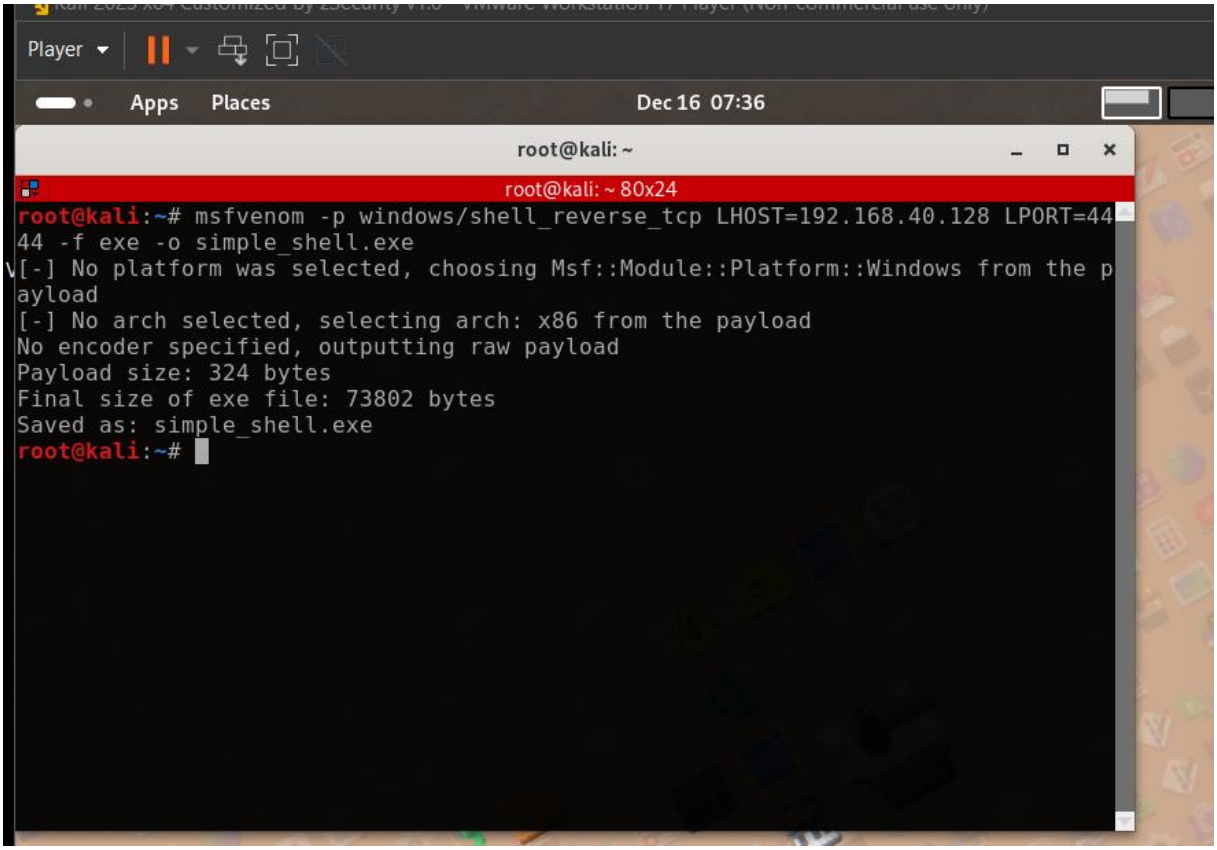

Figure 6.1:Reverse shell payload generation

3) A local HTTP server was started on the Kali machine to host the payload file.
4) The Windows 10 machine accessed the HTTP server and downloaded the payload.
5) The Metasploit Framework was launched on the Kali system.
6) A multi/handler listener was configured with the same payload, IP address, and port.
7) The listener was started and waited for incoming connections.
8) The payload was executed on the Windows 10 machine.

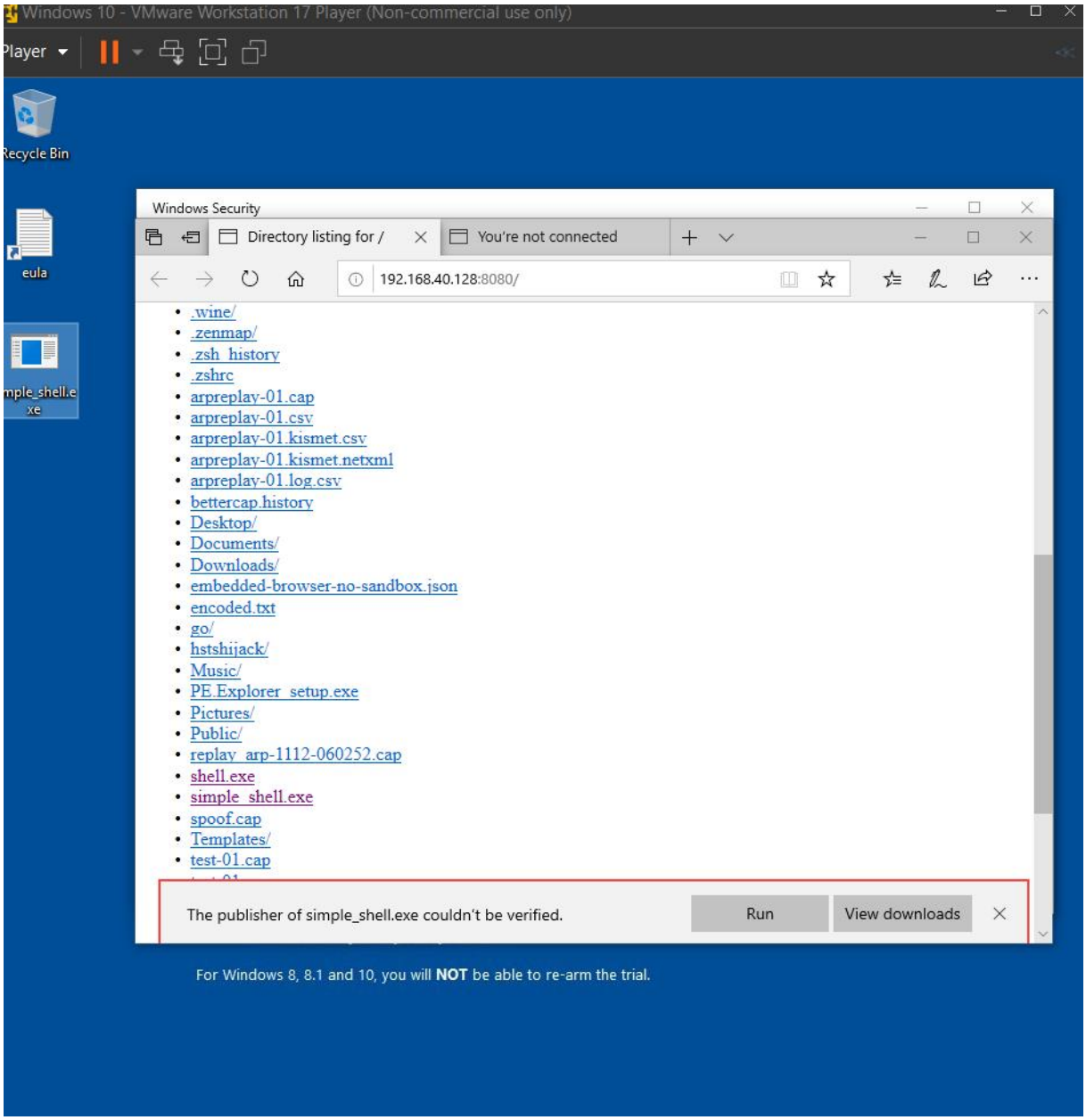

Figure 6.2:Payload execution on windows system

9) A successful reverse TCP connection was established from the Windows machine to the Kali machine.

10) A command shell session was opened, confirming remote access to the target system.

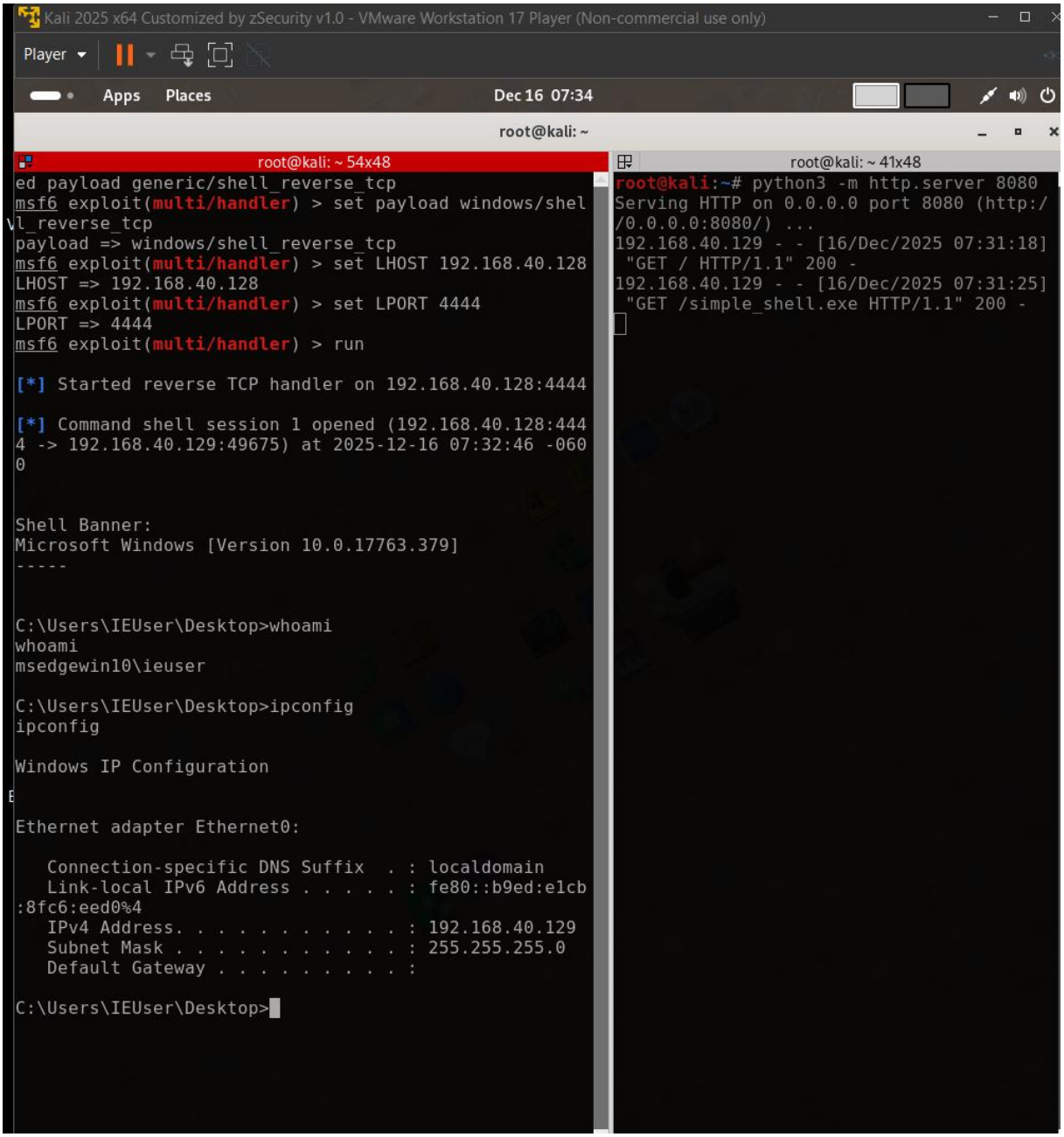

Figure 6.3:Metasploit listener and active session

➢ Analysis:-

- The reverse shell allowed the attacker system to gain remote command-line access to the Windows machine.
- The connection was initiated from the victim to the attacker, bypassing inbound firewall restrictions.
- Successful execution confirms that the target system allowed execution of untrusted binaries.

- The attacker was able to execute system commands such as whoami and ipconfig.
- This demonstrates a complete compromise of the target system at the user privilege level.

➢ Impact

Advantages (Security Learning Perspective)

1. Demonstrates real-world exploitation techniques
2. Helps understand how attackers gain remote access
3. Useful for penetration testing and red-team training
4. Improves understanding of payloads and listeners

Disadvantages (Security Risks)

5. Unauthorized remote access to the victim system
6. Possible data theft or system manipulation
7. Can lead to privilege escalation attacks
8. Full system compromise if combined with other exploits

➢ Mitigation Steps:-

- Disable execution of unknown or untrusted executables
- Use updated antivirus and endpoint protection solutions
- Restrict outbound network connections using firewalls
- Apply application whitelisting policies
- Regularly patch operating systems and software
- Monitor network traffic for suspicious outbound connections
- Educate users about malicious files and social engineering

➢ Resources Used:-

✧ **Attacker System:** Kali Linux
✧ **Target System:** Windows 10 Virtual Machine
✧ **Tools Used:**
✧ -Metasploit Framework
✧ -msfvenom
✧ -Python HTTP Server
✧ **Payload Type:** Windows Reverse TCP Shell

TASK 7: Wi-Fi Deauthentication Attack and WPA/WPA2 Handshake Analysis

➢ Attack Name:-

Wi-Fi Deauthentication Attack with WPA/WPA2 Handshake Capture and Offline Password Analysis

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➢ Severity:-

CVSS Score: 7.6

Severity Level: High

➢ Steps Performed:-

1) The wireless interface on the Kali Linux system was identified and prepared for monitoring.
2) Conflicting network processes were stopped to allow exclusive control of the wireless adapter.
3) The wireless adapter was configured to operate in monitor mode where by using a tool calledairmon-ng the managed mode is set to monitor mode which is necessary for wifi password cracking.

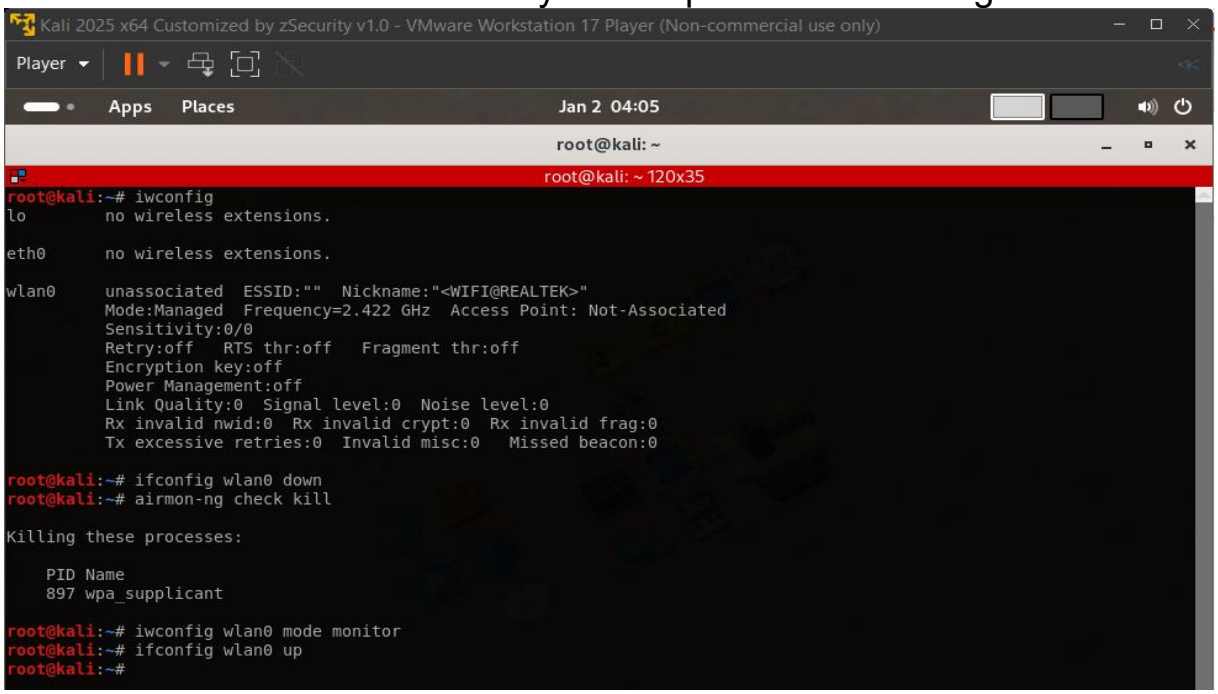

Figure 7.1:Monitor mode enabled

4) A passive scan was conducted to discover nearby wireless networks and connected client devices usind airodump-ng.The target Wi-Fi network (SSID, BSSID, channel, and encryption type) was identified which was oppo f17 pro.

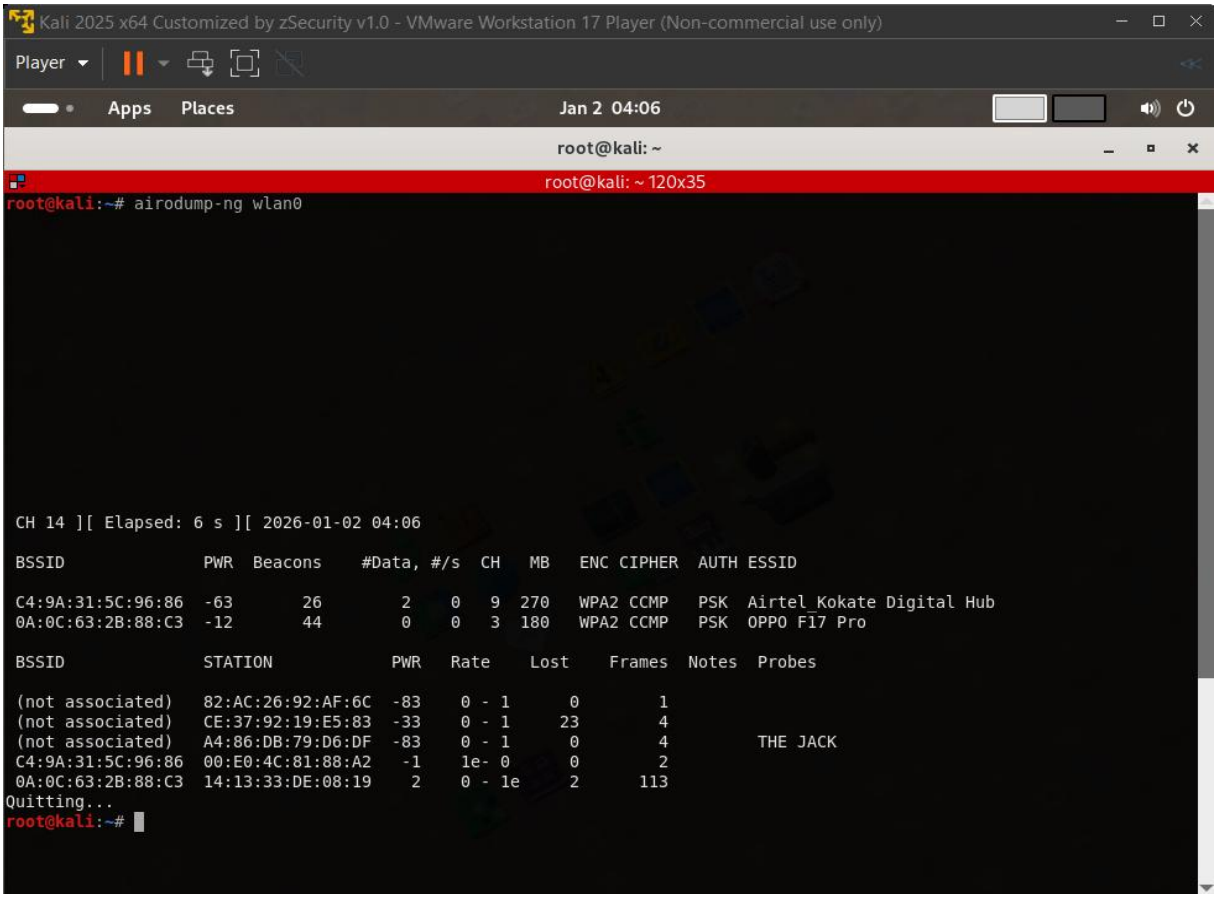

Figure 7.2:Nearby wireless networks detected

5) Packet capture was focused on the target network channel to collect authentication data.

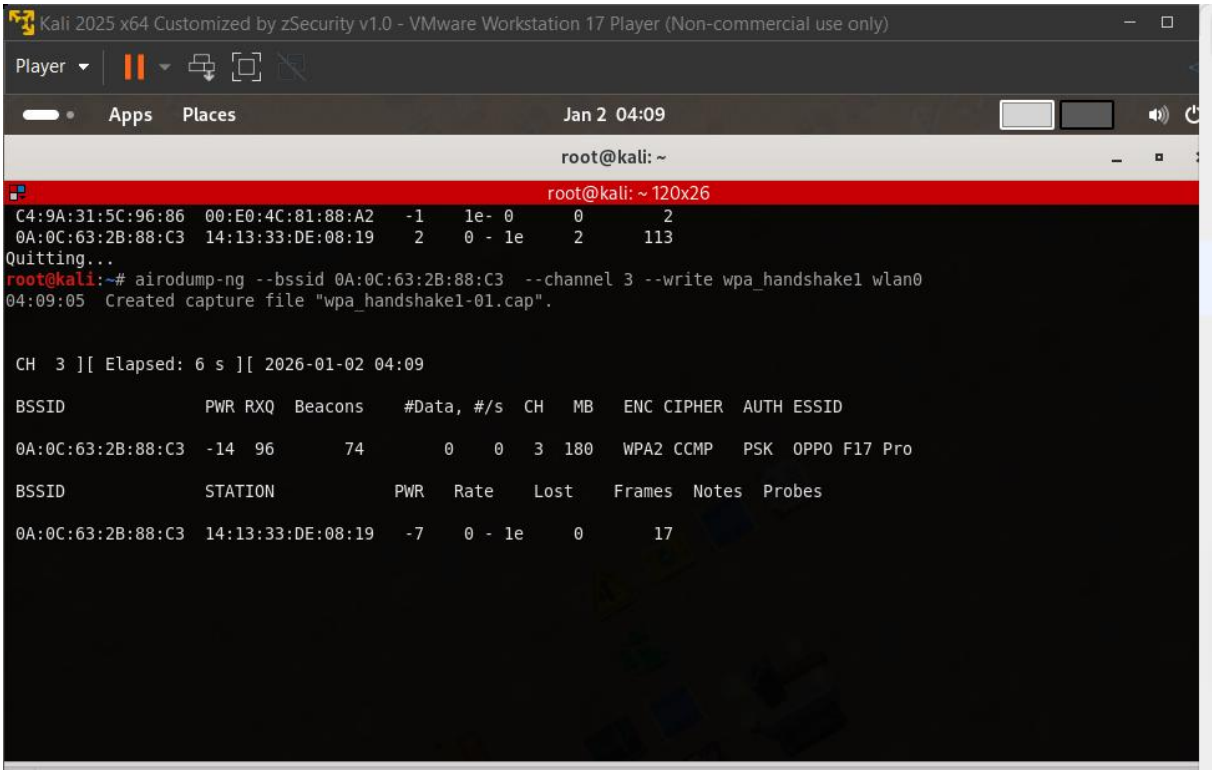

Figure 7.3:Handshake capture in progress

6) A deauthentication technique was used to temporarily disconnect a client device, forcing reconnection.Then WPA/WPA2 handshake was successfully captured during client activity and saved to a capture file named as wpa_handshake1-01.cap.

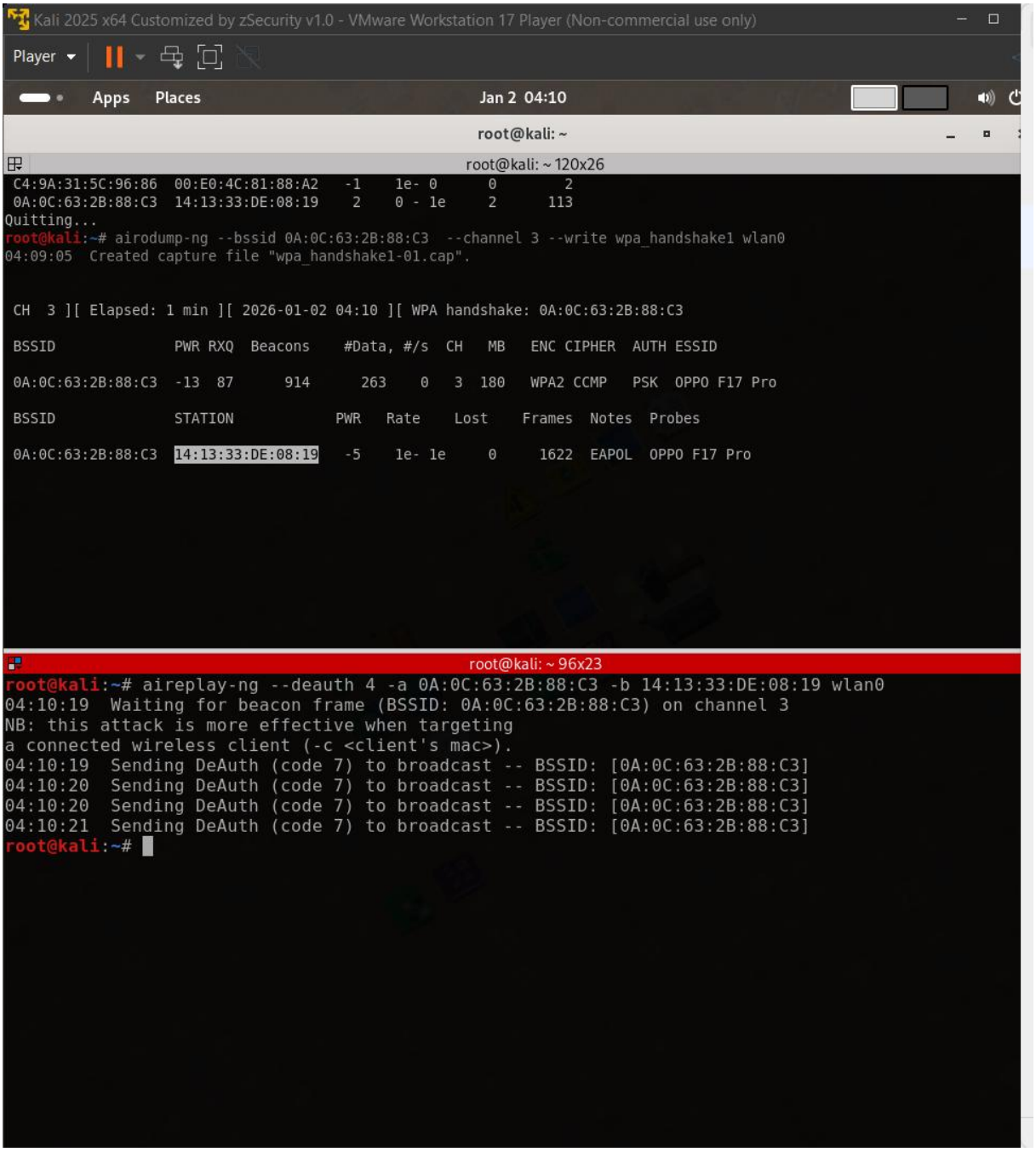

Figure 7.4:Deauthentication frame sent and WPA Handshake captured

7) The reconnection process generated EAPOL frames, confirming a valid handshake capture.
8) The captured handshake file was analyzed offline using a wordlist rockyou.txt and a tool called aircrack-ng.The Wi-Fi passphrase was successfully recovered during offline analysis and the key has been found as 1234509876.

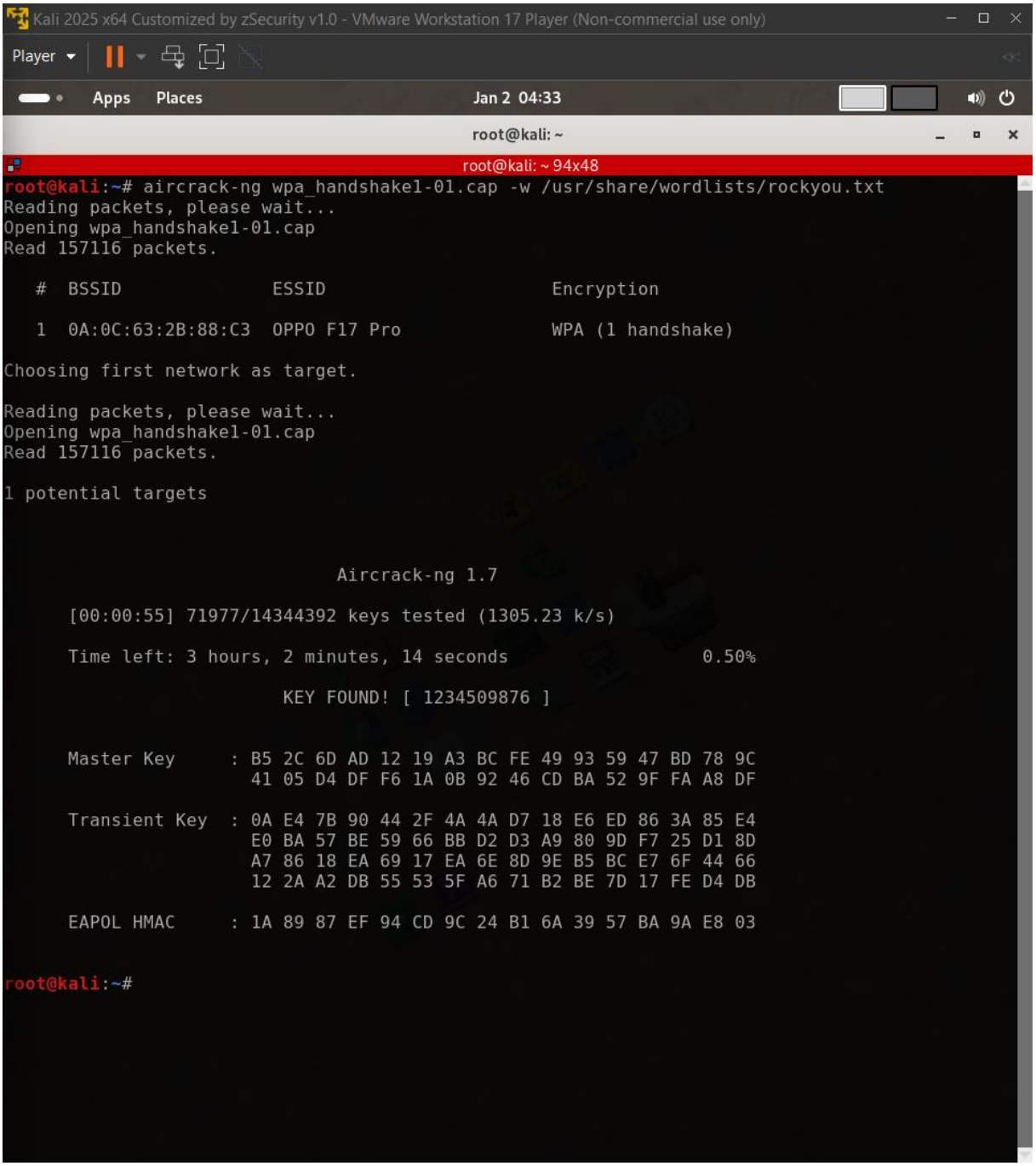

Figure 7.5:Offline password analysis result

➢ Analysis:-

- The screenshots confirm that the network used **WPA2-PSK with CCMP**, which is cryptographically strong.
- The attack did **not break encryption**; instead, it relied on **offline password guessing** using a captured handshake.
- The presence of a connected client enabled handshake capture during reconnection.
- Successful password recovery indicates the use of a **weak or guessable passphrase**.

- This demonstrates that **wireless security strength depends heavily on password complexity**, even when strong encryption is enabled.

➢ **Impact:-**

Advantages (Security Assessment Perspective)

1. Helps evaluate the real-world strength of Wi-Fi passwords
2. Demonstrates risks associated with weak passphrases
3. Useful for auditing personal or organizational wireless networks
4. Improves understanding of WPA/WPA2 authentication behavior

Disadvantages (Security Risks)

5. Unauthorized access to the wireless network
6. Potential misuse of network resources
7. Risk of internal network attacks after access is gained
8. Temporary service disruption during deauthentication

➢ **Mitigation Steps:-**

- Use **strong, long, and unique Wi-Fi passwords** (12–16+ characters)
- Enable **WPA3** or **WPA2-AES (CCMP)** only
- **Disable WPS** on the router-
- Monitor for abnormal deauthentication activity
- Limit Wi-Fi signal leakage beyond physical premises
- Regularly rotate Wi-Fi credentials
- Maintain updated router firmware

➢ **Resources Used:-**

✧ **Operating System:** Kali Linux
✧ **Hardware:** Wireless adapter supporting monitor mode
✧ **Tools Category:** Wireless auditing tools (authorized use only)
✧ **Encryption Tested:** WPA2-PSK (CCMP)
✧ **Wordlist Used:** rockyou.txt (offline analysis)
✧ **Target:** User-owned and authorized Wi-Fi network